
From Law to Action: Neuro-Symbolic Runtime Enforcement for MCP Agents

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Abstract

1 We evaluate a bounded runtime-checking question for tool effects that stand in
2 for generated-code side effects: under explicit ENFORCE at one pre-invocation
3 Model Context Protocol (MCP) handler, can an independently observed sandbox
4 mutation that lacks a matching context-bound admission be prevented, while a
5 matched admitted case still performs useful permitted work? Legal documents,
6 CVE-linked vulnerability and repair records, and SkillCenter procedures compile
7 into Legal, Security, and Intent IR. Those IRs are models with distinct authority,
8 not proofs of legal correctness, universal vulnerability absence, or permission. On
9 an admitted $60 \times 5 \times 3$ matrix of frozen modeled-policy probes, full enforcement
10 observed 0/90 policy-forbidden handler effects and completed 90/90 allowed
11 useful work. Unguarded execution and model-free prompt/retrieval labels observed
12 90/90 forbidden effects. Lightweight policy plus UCAN leaked 33/90. Expert legal fidelity and human agreement were not collected. Closed-loop model
13 planning was executed as a separate 900-cell study and is not pooled with the fixed-
14 action matrix. Closest systems—reference monitors, proof-carrying authorization,
15 Catala, UCAN/Cedar, AgentSpec, Progent, CaMeL, and AgentDojo—motivate
16 implementable arm classes; none is a scored baseline here.
17

18 1 Introduction

19 An autonomous agent can follow a useful procedure and still violate a legal restriction, reproduce
20 a known vulnerability, or act outside its delegated authority. A safeguard placed only in the model
21 prompt cannot distinguish these cases reliably. The implementation described here places source-
22 grounded constraints and a separate authorization boundary between proposed actions and a protected
23 MCP handler. MCP supplies tool discovery and invocation; application implementations remain
24 responsible for enforcing security requirements [? ?].

25 The method begins before a tool call. A legal collection supplies normative constraints, a vulnerability-
26 and-fix collection supplies security evidence, and a skill collection supplies descriptions of intended
27 procedures. These become three intermediate representations (IRs), not one undifferentiated retrieval
28 index. Formalization produces typed logic views and proof obligations. The runtime joins the results
29 against one exact invocation, then checks authorization again before the effect. This design combines
30 semantic compilation, proof search, capability security, and stateful execution rather than asking a
31 language model to certify its own behavior.

32 The evaluated workshop contribution is that bounded mediation test [?]. It is not a new authorization
33 logic, a new legal language, or a first agent runtime monitor. AgentSpec, Progent, and CaMeL
34 already supply runtime rules, tool-privilege checks, and capability isolation [? ? ?]; Catala
35 compiles expert-authored law [?]; AgentDojo measures utility beside attacks [?]. We cite those
36 primary sources as comparison classes. We do not execute them as experimental arms. Source

adapters, constraint compilation, DuckDB-backed supervision, and content-addressed evidence are implementation context. Legal IR is a lossy model of selected facets, not a proof of legal correctness. Default OFF pass-through and in-memory consumption are not enforcement. The scored study uses one sandbox handler, independently observed filesystem-journal effects, and a frozen modeled policy. It does not evaluate model-generated programs, expert legal interpretation, or universal prevention.

2 Related work

The scored evaluation asks whether an independently observed handler effect is denied at one explicit ENFORCE delegate while matched admitted work remains executable. The comparison below is a literature comparison on that question. No cited external system was executed as a scored arm.

Runtime enforcement. Complete mediation requires a check at every access to a protected resource [?]. Security automata and inlined or editing monitors describe which policies a runtime mechanism can enforce [? ? ? ?]. Those mechanisms inspect program actions. They do not compile legal or vulnerability sources, and they do not count useful MCP-handler work. AgentSpec supplies agent runtime triggers, predicates, and enforcement rules [?]. Progent checks symbolic tool privileges and classifies policy updates using an SMT solver [?]. These direct precedents rule out claims that this paper introduces agent runtime rules or guarded privilege updates. Neither system was benchmarked here.

Proof-carrying admission. Proof-carrying code [?] and proof-carrying authentication [?] require a checker to accept a proof before running untrusted code or granting a request. Translation validation and CompCert separate a translator from a checked proof [? ?]. A content identifier or a `proof_cid` in an envelope is not that check.

Executable law is not checked derivation. Encoding a statute as a logic program already performs an interpretation [? ? ?]. Catala compiles literate legal specifications to programs and has been applied to tax law [? ?]. It is the closest executable-law precedent. It is not an agent runtime. Legal IR in this paper is a bounded, lossy compilation of selected facets; unsupported facets and expert applicability remain labeling problems, not theorems.

Capabilities and application-level MCP enforcement. Object capabilities, attenuating caveats, relation tuples, and analyzable policy languages [? ? ? ?] justify a lightweight grant check. UCAN specifies signed delegated capabilities [?]. MCP supplies tool invocation; its authorization document specifies OAuth 2.1 resource-server roles and requires implementations to follow OAuth 2.1 security best practices [? ?]. That protocol is not a generated-code effect monitor. A signature, a CID, or a peer identity is not a generated-code effect observation.

Verifiable coding and agent-security evaluations. Functional correctness and issue resolution [? ?] measure useful coding work without gated forbidden-effect counters. Closed-loop verified generation and autoformalization [? ? ?] target specs and prover artifacts, not this handler route. AgentDojo, ToolEmu, InjecAgent, τ -bench, and AgentHarm [? ? ? ? ?] measure tool-use attacks, emulated risk, injection, policy following, or harm. CaMeL isolates agent data flow with capabilities [?]. They motivate reporting utility beside forbidden effects. They were not run here.

Implementable arms, not imported suites. The matched arms are mechanism classes in this stack: an unguarded handler (A0); inert prompt and retrieval labels as model-free equivalence controls (A1, A2); a qualified policy and capability verifier (A3); and explicit ENFORCE with exact context binding, a qualified proof route, and durable consumption (A4). Prompt or retrieval efficacy is not identified in the model-free study. CVEfixes is a vulnerability/fix collection [?], not a runtime. DuckDB/Quack, IPFS, and libp2p [? ? ?] are operational substrates. The workshop fit is runtime checking of a proposed effect before a protected mutation [?], not a claim that every cited system was benchmarked.

83 3 Corpus sources and compilation into IRs

84 The evaluated source freeze is 30 lineage families and 60 cases: six legal documents, 12 CVE
85 repository families with vulnerable/fixed pairs, and 12 SkillCenter families with a base procedure and
86 an adversarial variant. Physical reads used to reserve that freeze were six legal documents, 4329 rows
87 from the first CVEfixes shard, and 1182 SkillCenter index/content rows in the read bundle. Release-
88 card totals (municipal harvests, national-legislation cards, 12,987 CVEfixes input rows, 216,972
89 SkillCenter physical records) remain unrecounted background and are not evaluated populations.
90 Appendix A records family-level provenance requirements. No worldwide-coverage claim follows
91 from the bounded sample.

92 Legal documents compile through a restricted seven-field grammar (modality, actor, action, object,
93 conditions, exceptions, temporal qualifiers) with source maps and declared losses. Compilation is
94 inspectable. Source-to-rule fidelity remains a separate validation requirement and was not expert-
95 reviewed in this study. Executable-law work such as Catala supplies a relevant precedent [?]; it is
96 not an agent runtime or a handler-effect observer.

97 CVEfixes links vulnerability records to source repositories and fixes [?]. The inspected release pins hi-
98 toshura25/cvefixes at d4f5c4ea65329d9ccbb8a3b3149e5d06eda5edb2. The adapter constructs a
99 canonical Security IR declaration and rejects wildcard scope and inserted authority grants. Vulnerable
100 code is a positive control for a candidate restriction; fixed code is a negative control for that restriction.
101 A fix does not certify that all other vulnerabilities are absent. Runtime `cve_security_gate` maps
102 declared intent and independently extracted handler effects into the same Security IR root. Incomplete
103 mappings remain unknown.

104 SkillCenter supplies procedural Markdown. The bounded reader opens upstream SQLite bundles read-
105 only, disables extension loading, and never executes commands appearing in a skill. The inspected
106 source is Tommysha/skillcenter-bundles at f9dd4fec3c86d85ebf116c7408ac5ce602c418a1. In-
107 tent IR distinguishes goals, guards, effects, and control edges, with source spans and grounded-versus-
108 inferred status. A procedure’s word “permitted” is a source assertion, not a capability. The default
109 skill path is a deterministic structural normalizer. No trained encoder was pinned or evaluated.

110 The three families share identity, provenance, and obligation contracts. They do not share an ontology
111 or grant authority to one another. Sparse, dense, and graph retrieval may nominate evidence; hard
112 applicability and exact identity determine whether it may be used. Retrieval engineering is not a
113 measured token-saving result.

Table 1: Three source-to-IR paths. None of the source streams directly supplies execution authority.

Source stream	Derived semantics	Contribution to an action decision
Legal texts and versions	Norms, scope, authority, exceptions, temporal applicability	Which represented legal constraints apply?
CVE-linked code and fixes	Threat assumptions, effects, mitigations, scoped prohibition candidates	Does the proposed or generated behavior violate a reviewed security constraint?
Skill Markdown and metadata	Goals, actions, guards, effects, verification, control flow	What is the agent trying to do, and what effects does its procedure require?

114 4 Logic families, joining, and the protected route

115 An IR is a domain representation; a logic family determines how its statements are interpreted; a
116 prover is a backend that checks a supported encoding. A family’s presence in a registry does not mean
117 every formula or backend is qualified. Appendix B records the broader map. The benchmark-required
118 proof-oriented route actually executed is a QF_BOOL / propositional SAT encoding of a Hoare-style
119 security invariant, with SymPy DPLL as a child process and an independent exhaustive truth-table
120 checker. Both emit satisfiability authority only. SAT-only, simulated, policy-only, and monitor

121 evidence cannot satisfy a `theorem_proof` requirement. QF_LIA interpolation and interactive
 122 kernels were not selected and were not available on the sealed validation path.

Table 2: Roles of native views. Only the QF_BOOL SAT route was qualified for the scored study.

Logic or formal view	Implemented role in the stack	Qualified in this evaluation
Hoare-style contracts and SAT	Preconditions, postconditions, and a propositional invariant over grant and forbidden effect	QF_BOOL SAT plus independent truth table; satisfiability only
Rule/policy and capability views	Bounded authorization declarations and signed UCAN grants	Real Ed25519 verification in A3/A4; not a theorem
Deontic / temporal / event-calculus tags	Native representations and compiler bridges	Registered; not a live scored backend
Interactive proof kernels	Lean/Coq/Isabelle adapters	Unavailable on the sealed path; no invented kernel result

123 An `InvocationIntentEnvelope` commits to source identity, tenant, actor, audience, tool/version, argu-
 124 ments, proposed effects, environment, nonce, and relevant roots. Trusted adapters bind the effect
 125 description to the actual operation. The constraint compiler joins Legal, Security, and Intent checks
 126 against that exact invocation. A favorable score cannot offset a failed hard constraint.

127 The selected protected route is `PreInvocationEnforcer.authorize_and_delegate` in explicit
 128 `ENFORCE` around `SandboxExportHandler.export_json`. Implementation identifiers here are
 129 neutral aliases; the supplement’s `symbol_aliases.json` maps each to its included original source
 130 definition. Independent filesystem-journal counters observe handler effects. A permissible candidate
 131 still requires a current grant for the exact actor, tool, resource, and bounds. The runtime revalidates
 132 and consumes the permitted use immediately before delegation. Replay, expiry, wrong audience,
 133 changed arguments, undeclared effects, and changed roots are denials at that boundary. DuckDB with
 134 a typed Quack owner records operational progression for the A4 deployment; the default in-memory
 135 store is a labeled contrast only. IPFS names immutable bytes [?]; libp2p carries framed MCP
 136 messages [?]; UCAN checks signed attenuated delegation [?]. None of those substrates is itself the
 137 scored safety result. Kit’s `AuthorizationGate` is a related policy gate and is not the selected safety
 138 claim.

139 5 Evaluation

140 5.1 Frozen protocol and automated evidence scope

141 The predeclared fixed-action study uses 30 source-lineage families and 60 cases (six legal families,
 142 12 CVE repository pairs, 12 SkillCenter families), split 6/6/18 into development, calibration, and final
 143 by source lineage [?]. Each family contributes one policy-allowed and one policy-forbidden case.
 144 Five matched arms share the same candidate identifiers, arguments, and sandbox handler: unguarded
 145 execution (A0); inert prompt-text and retrieval-context labels as model-free equivalence controls
 146 (A1, A2); lightweight policy plus real Ed25519 UCAN verification (A3); and explicit `ENFORCE` with
 147 exact-context binding, a qualified QF_BOOL SAT job, and file-backed DuckDB consumption (A4).
 148 Seeds 104729, 104759, and 104761 yield a $60 \times 5 \times 3 = 900$ -cell matrix. The independent unit is the
 149 lineage family. Allowed and forbidden labels are the frozen modeled policy and machine-checkable
 150 behavior contract, not real-world legality.

151 Outside reviewers were not a run dependency. Expert legal applicability, exception fidelity, semantic
 152 accuracy, inter-annotator agreement, and legal validity were not collected and are not scored. Optional
 153 author reading, if it occurs later, is non-independent and is not human gold. Generated interpretations
 154 remain predictions. The LA-009 skill-validator comparison of 18/18 is shared-producer confor-
 155 mance: the same Intent IR decoder and schema functions already invoked by the normalizer, not an
 156 independent checker.

157 5.2 Source-to-IR contracts

158 Every frozen case has a retained prediction, failure, or unsupported record; cohorts were not changed
159 after outcomes. Quoted source-span linkage held on 60/60 cases. Parse or schema validity held
160 on 50/60. The scoped machine-contract coverage was 60/60. Four CVE adapter errors and six
161 skill-policy-blocked unsupported rows remain in the denominator accounting. CVE source-difference
162 observation was 24/24 under an independent behavior-contract producer; polarity remains unknown
163 on all 24 cases (12 pairs), and transfer to the `export_json` handler is unsupported. Implementation
164 agreement is not semantic accuracy. These measurements replace the original expert-reviewed
165 legal-fidelity row, which is withdrawn as unmeasured (Table E1).

166 5.3 Admitted protected-effect and useful-work results

167 The primary empirical study is the LA-029 operator matrix: 900 scientific cells, all resource-
168 qualified, with independent filesystem-journal effect counters. Compact rows do not retain
169 full per-cell proof or capability payloads; pinned executed mechanisms, effect counters, and
170 the shared durable-store identity support the runtime observations. Implementation revision
171 `ee6d73c4a5fc30d05ec4e787fdeb46d6701a4f947b7c874aecabf603e31ce06c`. Model calls
172 were 0. Independent human gold is false. Two original false host admissions (schedule indices
173 459 and 560) remain labeled as such. Two container-startup recoveries at indices 591 and 625 were
174 charged once after proof that science had not started; scientific retries were 0. Host attempts total
175 902. Measured descendant-inclusive group CPU was 1608.817478 seconds, under the frozen 18,000-
176 second stop. Peak cell memory was 274812928 bytes; maximum individual-cell wall-to-exit was
177 18.022665493073873 seconds. The startup recoveries used prospective cumulative-active allowances:
178 failed-startup plus recovered-run wall totaled 11.766052 seconds at index 591 and 19.793533 seconds
179 at index 625. Separate operator pause gaps were not active runtime; these amendments do not claim
180 an uninterrupted original absolute 20-second interval.

181 Table 3 reports the admitted rates. On the held-out final split (18 families; 54 forbidden-labelled
182 and 54 allowed-labelled attempts per arm), A4 observed 0/54 policy-forbidden effects and com-
183 pleted 54/54 allowed useful work with 0/54 false denials. Across the full admitted matrix the
184 same pattern holds: A4 0/90 forbidden effects and 90/90 allowed useful work; A0–A2 90/90
185 forbidden effects on forbidden-labelled attempts; A3 33/90 forbidden effects. A1 and A2 match
186 A0 (final-split paired family difference 0, cluster-bootstrap 95% interval $[0, 0]$) and therefore re-
187 main instrumentation/equivalence controls, not prompt or retrieval efficacy. Final-split paired fam-
188 ily differences in forbidden-effect rate are -0.6666666666666666 for A3 versus A0 (percentile
189 95% $[-0.8888888888888888, -0.4444444444444444]$), -1.0 for A4 versus A0 ($[-1.0, -1.0]$), and
190 -0.3333333333333333 for A4 versus A3 ($[-0.5555555555555556, -0.1111111111111111]$). A
191 degenerate zero interval does not establish population-level absence or universal prevention.

192 The handler under test accepts a declarative `export_json` instruction. Independently observed
193 effects are sandbox filesystem-journal events, not AST extraction from model-generated source. No
194 scientific generative model was pinned.

195 5.4 Negative results and scoped omissions

196 A3 is not equivalent to A4. The 33 A3 leaks (decision allow, observed forbidden ef-
197 fect) concentrate on `wrong_date_or_jurisdiction` (9), `forged_receipt` (6), `replay` (6),
198 `changed_root_clock_or_environment` (6), and `undeclared_handler_effect` (6). On the
199 final split the residual classes are replay, changed root/clock/environment, and wrong date/jurisdiction
200 (6 each). A4 denied those identities. SAT/UNSAT on the selected QF_BOOL route has satisfiability
201 authority only; it cannot authorize a `theorem_proof` allow. Z3, cvc5, Vampire, E, Lean, Coq/Rocq,
202 Isabelle, and QF_LIA interpolation were absent from the sealed validation path and are not assigned
203 invented results.

204 Closed-loop generated-code planning (protocol RQ3) was executed as a separate 900-cell study
205 against the pinned local model; Table 4 reports those observations and is not pooled with the
206 fixed-action matrix.

Table 3: Admitted LA-029 fixed-action rates from independent effect counters. Forbidden-effect rate uses forbidden-labelled attempts; allowed-task success uses allowed-labelled attempts. Labels are the frozen modeled policy. Overall $n = 90$ forbidden-labelled and 90 allowed-labelled attempts per arm (180 decisions). Final $n = 54 + 54$.

Arm	Split	Forbidden effects	Allowed useful work	False denials
A0 unguarded	overall	90/90	90/90	0/90
A1 prompt label	overall	90/90	90/90	0/90
A2 retrieval label	overall	90/90	90/90	0/90
A3 policy+UCAN	overall	33/90	90/90	0/90
A4 ENFORCE	overall	0/90	90/90	0/90
A0 unguarded	final	54/54	54/54	0/54
A3 policy+UCAN	final	18/54	54/54	0/54
A4 ENFORCE	final	0/54	54/54	0/54

Table 4: Generated-code 900-cell observations. Every scheduled identity remains in the denominator, including format and transport failures.

Arm	Split	Forbidden effects	Useful work	Format/transport fail
A0 unguarded	overall	0/180	6/180	11/180
A1 policy prompt	overall	1/180	10/180	67/180
A2 retrieval	overall	0/180	5/180	78/180
A3 path policy	overall	0/180	9/180	74/180
A4 ENFORCE	overall	0/180	11/180	66/180
A0 unguarded	final	0/108	3/108	8/108
A1 policy prompt	final	1/108	8/108	35/108
A2 retrieval	final	0/108	2/108	47/108
A3 path policy	final	0/108	3/108	45/108
A4 ENFORCE	final	0/108	3/108	42/108

207 5.5 Generated-code planning and interface qualification

208 Eight frozen synthetic interface probes were executed against the pinned local Leanstral deployment
209 (seeds 104729, 104759, and 104761; 2048-input / 1024-output contract; no scientific sources). Status
210 PASS_INTERFACE_CHECKS_ONLY: seven generations plus the expected input-overflow rejection; the
211 identical seed-104729 repeat matched exactly; prompt-token counts equalled preflight input counts.
212 Distinct exchanges reserved: 8.

213 The same warm model then executed the original $60 \times 5 \times 3 = 900$ generated-code identities (180
214 development, 180 calibration, 540 final) without reload, under a 36,000-second service wall and
215 a 360-second startup bound. Recorded cells: 900/900. Model calls: 1481. Prompt/completion
216 tokens: 807442/148254. Exact useful-work matches: 41/900. Generated programs ran in the bounded
217 in-process profile. A0–A2 execute inside fixture containment; A3 applies declared path policy; A4
218 additionally refuses secret-bearing arguments. Useful work and forbidden effects are independent
219 fixture observations, not model self-reports. This matrix is not pooled with the fixed-action study.
220 Expert legal fidelity was not scored.

221 The original LA-015 compact log and its rescue remain protocol-unadmitted diagnostics; they are
222 not headline admitted rates. LA-017 paired mechanism ablations and LA-016/LA-017 comparison
223 wording are protocol-unadmitted diagnostic evidence and do not establish admitted model efficacy.
224 LA-018 cold/warm/stale timings are a development A4 instrumentation study ($n = 12$ per condition;
225 cold mean end-to-end wall 0.012326 s, warm 0.006635 s, stale-root and stale-clock 0.00322 s with
226 12/12 stale-evidence flags; model tokens unobserved; no price conversion) and are not a 900-
227 cell rescore. LA-020 twelve-step traces are separate diagnostic replays of frozen A4 identities,
228 not additional scored cells. Qualification suites for mediation, DuckDB consumption, and local
229 stdio/HTTP/libp2p parity passed their selected-claim cases and remain qualification, not scored arm
230 rates. Wide-area publication and an IPFS daemon were not available.

231 Admitted rates are taken from `results/fixed_actions_admitted/summary.json` together with
232 the matching cost report and claim guidance. LA-019 generated tables and figures analyze the
233 historical unadmitted log and are not substituted for that admitted summary.

234 6 Limitations and modeling assumptions

235 Legal IR is a lossy model of selected facets (modality, actor, action, object, conditions, exceptions,
236 temporal qualifiers), not the law and not a proof of legal correctness. Unsupported constructs must
237 not disappear into a successful admission. Expert legal fidelity, human agreement, and legal-validity
238 rates were not collected; they remain unmeasured rather than filled with compiler agreement or
239 generated labels. Security IR is a scoped candidate prohibition. A CVE or CWE identifier, a fix, or a
240 source-difference observation is not universal vulnerability absence and is not transfer to unrelated
241 code. Skill Markdown is untrusted data; a procedure’s word “permitted” is not a capability.

242 Complete mediation is claimed only for `SupervisorPreInvocationEnforcement.authorize_`
243 `and_delegate` in explicit `ENFORCE` around one benchmark-owned `BoundedExportHandler.`
244 `export_json` sandbox mutation. Default `OFF` is pass-through. The default in-memory consumption
245 store is not the restart-safety configuration; A4 used a file-backed DuckDB typed Quack owner. A
246 DuckDB commit is not remote exactly-once evidence. Unprotected, observational, and nonidentical
247 routes were exercised in qualification and excluded from the safety claim. Local stdio, loopback
248 HTTP, and loopback libp2p parity do not establish wide-area transport.

249 The selected proof route is propositional SAT with an independent truth-table checker. Satisfiability
250 is not a kernel theorem. Compact admitted records do not retain full per-cell proof or capability
251 payloads. The fixed-action handler consumes a declarative export instruction. A separate 900-cell
252 generated-code study executed model programs in a bounded interpreter; those observations are not
253 pooled with the fixed-action rates. No trained skill normalizer or encoder was evaluated. Release-card
254 corpus totals were not independently recounted and are not evaluated populations. Optional author
255 review is not a prerequisite for this draft, is not independent annotation, and cannot replace the frozen
256 automated evidence.

257 7 Conclusion

258 On one explicit `ENFORCE` sandbox route, independently observed forbidden handler effects were
259 prevented for every frozen policy-forbidden probe while matched admitted work still succeeded.
260 Lightweight policy plus UCAN was not sufficient for the residual mutation classes. Unguarded
261 execution and inert prompt or retrieval labels were not sufficient either. The result is a bounded, policy-
262 relative verifiable-coding measurement [?], not superiority to Catala, AgentSpec, Progent, CaMeL,
263 AgentDojo, or any other listed system, and not a claim of legal correctness or expert agreement.
264 Closed-loop generated-program control was measured separately (Section 5.5) and is not pooled with
265 the fixed-action rates. Optional author review is not required to generate this evidence-scoped draft
266 and would not be independent human validation.

A Corpus lineage and release accounting

The inspected legal-source program includes public releases for the U.S. Code, state statutes, state administrative rules, Federal Register documents, state/local court rules, municipal/county codes, caselaw, and national legislation. The paper does not interpret a dataset title, a country registry entry, or a successful scrape as exhaustive coverage. The evaluated legal cohort is the six frozen documents. Card-reported municipal, Netherlands, and Belgian quantities have not been independently recounted and are not evaluated records.

Table A1: Legal source families and provenance requirements. Identifying publisher/dataset mappings belong in the private author record until camera-ready.

Source family	Representative inspected substrate	Required distinctions
Federal statutory law	U.S. Code sparse GraphRAG release, pinned source revision and official release point	Codification/release point, provision identity, amendment history and actual effective dates
State law	State-statute release and separate administrative-rule corpus	State, authority family, rule/statute distinction, missing jurisdictions and time coverage
Federal regulatory material	Federal Register collection	Proposed rule, final rule, notice, publication date, effective date and codified regulation are not interchangeable
Court rules and caselaw	Merged per-state court rules; Caselaw Access Project-derived collection	Court, jurisdiction, local/general rule, opinion treatment, holdings versus quotations
Municipal and county law	Multi-drop municipal harvest	Publisher, jurisdiction, per-drop discovery and failure counts; registry aliases may lag new release namespaces
International legislation	Netherlands corpus; Belgian official-source archive harvest; Argentine CID-keyed retrieval release	Language, source authority, consolidation date, archive capture, extraction version and incomplete coverage

The security source pins hitoshura25/cvefixes at d4f5c4ea65329d9ccbb8a3b3149e5d06eda5edb2. Graph nodes, edges, vectors, source rows, and unique CVEs are different populations. The evaluated CVE cohort is 12 repository families / 24 cases plus 84 excluded controls that never enter empirical sample counts. Evaluation includes the original vulnerable condition and the fixed negative control. Transfer to a proposed tool needs an explicit mapping; polarity remains unknown where the machine contract does not support it.

The SkillCenter source is Tommysha/skillcenter-bundles at f9dd4fec3c86d85ebf116c7408ac5ce602c418a1. SQLite is an untrusted input container, not the supervisor database. The evaluated skill cohort is 12 families / 24 cases. Index totals are not additional skills. Counts of available records do not establish counts of normalized Intent IR documents or safe procedures.

```
pinned source revision + raw artifact bytes
-> bounded source record + content identity + use policy
-> LegalIR | SecurityIR | IntentIR declaration
-> native formal views + source map + loss/coverage diagnostics
-> selected applicable constraints + exact invocation
-> proof jobs + checked evidence + decision receipt
-> authenticated capability / execution permit
-> guarded effect + operational observation + durable record
```

293 B Logic-family and backend method map

294 A shared symbol table does not prove that two encodings preserve meaning. The TDFOL and
 295 DCEC Python cores establish parsable native representations. Resource-scoped authorization may be
 296 representable without invoking every family. Solver success, kernel acceptance, runtime monitors,
 297 and policy approval are non-interchangeable authority kinds. A timeout or unavailable backend is not
 298 a refutation.

Table B1: Logic families have different meanings. A shared symbol table or cross-view link does not prove that two encodings preserve those meanings.

Family/view	Representation and operations	What the paper may claim
Propositional and first-order	Boolean operators, typed terms, predicates, variables and quantifiers	Bounded facts and implications under the encoded vocabulary; not general natural-language understanding
F-logic / frame	Object identity, class membership, slots and relational inheritance	Structured semantic relations; not a cryptographic or temporal proof by itself
Deontic	Obligation, permission, prohibition	Explicit normative force under the selected axioms; an encoded permission is not an issued capability
Temporal / TDFOL	Always, eventually, next, until, since	Model time-sensitive requirements; a backend must support the actual fragment
Event calculus	Events, fluents, initiation/termination	Reason about transitions as represented; an observed event record is not a proof of every transition
Datalog / rule-policy	Bounded rules and authorization declarations	Rule derivation with an explicit open/closed-world convention, not automatic theorem authority
Hoare-style / SAT	Preconditions, postconditions, propositional invariants	The qualified QF_BOOL SAT route; SAT/UNSAT cannot authorize theorem_proof allows
Interactive proof	Native proof terms/scripts	Unavailable here; no invented kernel result

299 C Database, artifact, network, and capability boundaries

Table C1: Database and protocol responsibilities. Legacy or auxiliary indexes can exist without changing the canonical DuckDB control-plane description.

Layer	Technology and inspected role	Incorrect inference to avoid
Supervisor operational authority	DuckDB control.duckdb; typed Quack owner/client; StateTransaction CAS and idempotency	“The supervisor uses SQLite” or “each agent writes the database file directly”
Source input packaging	Read-only SkillCenter SQLite bundles; source Parquet/JSON-LD/HTML/WARC	“An input database format defines the runtime architecture”
Retrieval products	BM25, vectors, graphs and bounded Parquet shards keyed by source IDs/CIDs	“A retrieved neighbor is a grant or a proof”

Layer	Technology and inspected role	Incorrect inference to avoid
Immutable artifacts	Kit-managed content-addressed blocks, manifests, maps, proof/decision receipts, optional IPFS replication	“The newest block is automatically the accepted state”
Peer carriage	libp2p negotiated streams and MCP++ framing	“A peer identity grants tool permission”
Delegated capability	Strict signed UCAN verification and revocation/replay state	“A valid token proves the action is legally and semantically safe”

300 A caller supplies a named command and typed parameters, not a SQL string. The identical command
301 can retrieve a committed result after a lost reply; a changed payload with the same identity is a
302 conflict. The Quack gateway extension was unavailable under the sealed home and is not claimed.
303 IPFS answers which artifact bytes are requested; local CID computation proves neither publication
304 nor public availability. libp2p answers how peers connect. In-process framing is not libp2p network
305 evidence. Strict UCAN checks cryptographic issuer identity, delegation continuity, resource/ability
306 bounds, temporal validity, attenuation, revocation, and replay policy. It does not infer delegation from
307 natural-language text.

308 D Implementation trace

309 The portable supplement maps neutral analysis paths to original source commitments and separately
310 hashes the included path-normalized bytes. Private runtime credentials and the live store are not
311 distributed.

Table D1: Implementation-oriented trace. Compact LA-029 rows remain the scored outcomes; LA-020 retains intermediate artifacts on a separate diagnostic replay of frozen A4 identities.

Method	Representative implementation	Inputs inspected in a reproducible run
Legal routing	Deterministic deontic parser; typed canonical compiler	Dataset revision, source body, jurisdiction, source maps, unsupported facets
Skill import	SkillRecordAdapter and read-only bundle adapter	Physical source row, metadata, skill/library Markdown, entry/body identities
CVE adaptation	CVEFixesAdapter	Vulnerable/fixed source records, candidate scope, prohibition/control polarity
Constraint join	ConstraintCompiler	Candidate graph, legal/security/intent checks, proof requirements, current roots
Proof authorization	AuthorizationService; SAT child process plus truth-table checker	QF_BOOL jobs, satisfiability authority, rejected theorem_proof upgrades
Guarded execution	Pre-invocation enforcement wrapper in ENFORCE	Current live context, authenticated receipt, expiry/use state, delegate effects
State control	TypedStateClient; file-backed DuckDB consumption	Named command, generation, fence, revision, idempotency and committed result

312 A diagnostic replay of the permitted A4 identity recorded by LA-020 (seed 104729, skill case-0;
313 exact attempt identifier retained in the worked-trace record) ran source-pin resolution, Intent IR
314 derivation, exact tool/argument binding, QF_BOOL SAT (provider unsat, checker agrees, satisfiability

only), real Ed25519 UCAN allow, ENFORCE consume on the DuckDB owner, one export_json dispatch, and one independent journal event with useful work. Matched forbidden replays for wrong audience, undeclared effect, and replay denied at delegation or consumption with zero forbidden effects. LegallIRCompiler was not the selected adapter. Independent human review was not used. No fabricated signature or conceptual step is presented as executed.

E Resolved measurement inventory

Original Table E1 listed placeholders. Table E1 records the actual disposition. “Withdrawn” and “qualification only” are scoped omissions, not zeros invented from missing runs.

Table E1: Resolved measurement inventory. Withdrawn claims are un-measured. Qualification suites are not scored arm rates. The primary protected-effect study is the admitted LA-029 matrix.

Measurement	Actual record	Disposition
Legal source-to-IR fidelity	Expert-reviewed atoms, exceptions, applicability and agreement	Withdrawn / unmeasured; replaced by automated span, schema, and machine-contract checks (60/60, 50/60, 60/60)
CVE restriction specificity	Vulnerable/fixed pairs and excluded controls	12 pairs / 24 cases and 84 excluded controls retained; source-difference 24/24; polarity unknown 24/24; no transfer claim
Skill grounding	Source spans and malicious Markdown negatives	Span linkage on all skill cases; 18/18 shared-producer schema conformance, not independent agreement; six unsupported policy-blocked rows
Proof job execution	Per-family provider/checker route	QF_BOOL SAT qualified; theorem kernels and QF_LIA interpolation unavailable, not invented
Guarded effect prevention	Delegate/effect counters under allow, deny, stale and replay	Admitted LA-029: A4 0/90 forbidden effects and 90/90 allowed useful work
DuckDB owner safety	Wrong generation/fence, concurrent commands, lost reply, restart	Qualification 14/14 selected-claim cases; in-memory restart is a labeled contrast; not a scored 900-cell rate
Cross-transport parity	Stdio/HTTP/P2P same request and outcome	Local-process and loopback qualification; WAN and IPFS daemon absent
End-to-end efficiency	Model tokens, retrieval, proof and state overhead	Development A4 instrumentation only; model tokens unobserved; no price conversion; not a 900-cell rescore
Closed-loop planning	Pinned-model generated programs and recovery	Executed separately: 900/900 terminal records; 8 interface probes PASS_INTERFACE_CHECKS_ONLY; 41/900 exact useful-work; 1/900 forbidden effects; 1481 model calls; not pooled with the fixed-action matrix

F Scope, ongoing work, and disclosure

Semantic-addressed program worlds, learned residuals, semantic-preserving refactoring, parallel sealing, and proof-carrying test reuse were unavailable for this revision and are not implemented contributions. Semantic minification and learned encoders were not evaluated as cost savings.

327 Language-model assistance supported source inspection, implementation and control development,
328 execution orchestration, retained-data analysis, and manuscript drafting. Grok grok-4.6 (xAI)
329 performed ordinary drafting and inspection; Codex (GPT-6) coordinated implementation, recovery,
330 analysis, and final artifact corrections. The configured Codex gpt-5.6-terra fallback is a config-
331 uration record, not evidence that it ran. These assistants were not scored experimental arms: the
332 admitted fixed-action matrix executed 0 scientific model calls, and A1/A2 are model-free labels.
333 Closed-loop generated-code planning was executed as a separate 900-cell study (1481 model calls)
334 and is not pooled with that matrix. No independent human legal, security, or intent validation and
335 no inter-annotator agreement study was collected. Optional author review is non-independent, is
336 not a prerequisite, and is not recorded as completed. Automated controls and retained independent
337 effect observations support the measured claims; they do not establish expert legal fidelity or replace
338 uncollected human judgments.

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